

2.1 The Hard Disk

2.1.1 Access Time

Reading data from or writing data to a hard disk involves the rotation of the spindle to bring the required sector under the read/write head, and the motion of the read / write head to the desired track on the platter. The access time is the sum of the time taken for these.

2.1.2 Active Partition

This is the partition that has been marked in the Master Partition Table as containing the boot information. Only one partition can be marked as active, though more than one can contain the boot information.

2.1.3 ATA / IDE

The ATA (Advanced Technology Attachment) specification deals with the interface between the motherboard and hard disks that have integrated circuitry. This standard is also referred to as IDE (or Integrated Drive Electronics). Earlier, the hard disk controller had to be separately plugged into an expansion slot, with cables running to the hard disk from the controller. ATA has many versions, which have increasingly improved the features and speeds of the connection. ATA 2 is also called Extended IDE (EIDE) because it introduced support for large drives by using the LBA naming scheme. ATA 6 is the last version of ATA, and is also referred to as ATA/133, which denotes the maximum data rate for the interface—133 Megabytes per second.

2.1.4 ATAPI

Advanced Technology Attachment Packet Interface is a protocol that allows optical drives to be connected through PATA cables to IDE controllers. This forms part of the ATA version 4 specifications.

2.1.5 Bad Sector

Bad sectors are areas on a hard disk that are physically damaged. The data present in these areas cannot be read with normal soft-